

Assignment 7&8

(Due on Jun. 14, 2026)

Question 1

Tungsten W ($\phi_m = 4.55 \text{ eV}$) is contacted to phosphorus (P)-doped silicon with a P concentration of $1 \times 10^{17} \text{ cm}^{-3}$ at room temperature. The electron affinity χ is 4.05 eV .

- Calculate ideal Schottky barrier height and built-in potential barrier in silicon. (10 pts)
- Plot electric field distribution across the metal-silicon junction when -3 V (with respect to silicon) is applied to metal. Where is the position of peak electric field? (15 pts)

Question 5

(a)

$$\phi_{B0} = \phi_m - \chi = 4.55 \text{ V} - 4.05 \text{ V} = 0.5 \text{ V}$$

$$\phi_n = \frac{kT}{e} \ln \left(\frac{N_c}{N_d} \right) = 0.0259 \text{ V} \times \ln \left(\frac{2.8 \times 10^{19} \text{ cm}^{-3}}{10^{17} \text{ cm}^{-3}} \right) = 0.146 \text{ V}$$

$$V_{bi} = \phi_{B0} - \phi_n = 0.5 \text{ V} - 0.146 \text{ V} = 0.354 \text{ V}$$

(b)

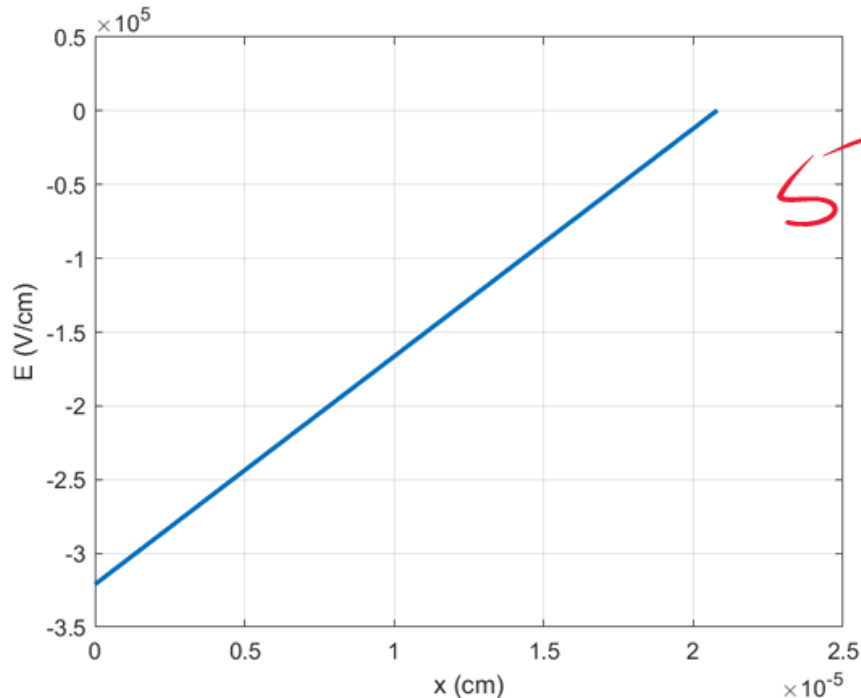
$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d}} = \sqrt{\frac{2(11.7)(8.85 \times 10^{-14} \text{ F/cm})(0.354 \text{ V} + 3 \text{ V})}{1.6 \times 10^{-19} \text{ C} \times 10^{17} \text{ cm}^{-3}}} = 2.08 \times 10^{-5} \text{ cm}$$

$$= 0.208 \mu\text{m}$$

$$E = -\frac{eN_d}{\epsilon_s}(x_n - x)$$

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.6 \times 10^{-19} \text{ C})(10^{17} \text{ cm}^{-3})(2.08 \times 10^{-5} \text{ cm})}{(11.7)(8.85 \times 10^{-14} \text{ F/cm})} = 3.21 \times 10^5 \text{ V/cm}$$

The electric field reached a peak value at the metal-semiconductor interface.



Question 2

Consider a MOS capacitor fabricated on a p-type wafer. The p-type substrate doping concentration is $1 \times 10^{16} \text{ cm}^{-3}$ and gate oxide thickness is 100 Å gate oxide. Phosphorus-doped polysilicon layer is used for gate electrode. Phosphorus doping concentration in the polysilicon gate is $1 \times 10^{20} \text{ cm}^{-3}$. The area of the capacitor is $500 \times 100 \text{ } \mu\text{m}^2$.

- Calculate the surface potential at the onset point of strong inversion. (6 pts)
- Calculate the minimum capacitance at high frequency C-V measurement. (5 pts)
- The flat-band voltage was found to be -1.5 V from the C-V measurement. What is the total number of fixed oxide charge? (6 pts)
- Calculate threshold voltage under condition (c). (8 pts)

Q6. (a). $\phi_b = \frac{kT}{e} \ln \frac{N_A}{n_i} = 0.0259 \ln \frac{10^{16}}{1.5 \times 10^{10}} = 0.347 \text{ V}$. $n_i = 6.8 \times 10^9$
 $\phi_s = 2\phi_b = 0.694 \text{ V}$. 0.734 V .

(b). $C_{\min} = C_{ox}$ 串联 $C_d \rightarrow$ 耗尽区宽度 $\frac{1}{2}$.

$C_{d\min}$ 在 $W_{d\max}$ 时达到. $C_{d\min} = \frac{\epsilon_s}{W_{d\max}}$.

$W_{d\max} = W_{\phi_s} = \sqrt{\frac{4\epsilon_s \phi_b}{e N_A}} = 3.00 \times 10^{-5} \text{ cm} / 3.08 \times 10^{-5} \text{ cm}$.

$C_{d\min} = \frac{\epsilon_s}{W_{d\max}} = 3.45 \times 10^{-8} \text{ F/cm}^2$. 取 $N_A = 1.5 \times 10^{16} \text{ cm}^{-3}$.

$C_{ox} = \frac{\epsilon_{SiO_2}}{t_{ox}} = \frac{3.9 \epsilon_0}{1 \times 10^{-6} \text{ cm}} = 5.45 \times 10^{-7} \text{ F/cm}^2$.

$\therefore C_{\min} = \frac{C_{d\min} \times C_{ox}}{C_{d\min} + C_{ox}} = 3.14 \times 10^{-8} \text{ F/cm}^2$.

$C_{\min} = C_{\min} \cdot A = 1.57 \times 10^{-11} \text{ F}$

(c). 非理想电势. $Q_{fix} = \Delta V_{FB} \cdot C_{ox} \cdot A$

$n^+ \text{ poly: } E_{fn} - E_i = kT \ln \frac{N_D}{n_i} = 0.59 \text{ V} > \frac{1}{2} E_g$. 认为 E_{fn} 在 E_c .

$p \text{ sub: } E_i - E_{fp} = kT \ln \frac{N_A}{n_i} = 0.347 \text{ V}$. 0.367 V . (ϕ_b)

$\therefore V_{FB \text{ ideal}} = \phi_{ms} = -\frac{1.12}{2} - \phi_b = -0.907 \text{ V} / -0.927 \text{ V}$.

$\therefore \Delta V_{FB} = -1.5 - (V_{FB \text{ ideal}}) = -0.593 \text{ V} / -0.573 \text{ V}$.

$Q_{fix} = \Delta V_{FB} \cdot C_{ox} \cdot A$

$= 1.02 \times 10^{-10} \text{ C} / 9.88 \times 10^{-11} \text{ C}$.

$N_{fix} = \frac{Q_{fix}}{e} = 6.37 \times 10^8 / 6.17 \times 10^8$

(d). $V_T = V_{FB} - \frac{Q_d}{C_{ox}} + 2\phi_b$

$= V_{FB} + \frac{\sqrt{e N_A \epsilon_s \phi_b}}{C_{ox}} + 2\phi_b$

$= -0.67 / 0.63 \text{ V}$

Question 3

- (a) An engineer was measuring the high-frequency C-V characteristic of a MOS capacitor with a p-type substrate and 100 Å thick gate oxide. In order to measure the **maximum depletion region width** in the inversion mode (threshold voltage = 2 V), he swept the gate voltage, V_G , from -4 V to +4 V. At $V_G = +4$ V, he recorded the capacitance value (C_1) while keeping the voltage constant. However, after reading the C_1 value, he found that the C_1 value was increasing with time and finally settled to another value C_2 . In this case, is his estimated maximum depletion width based on the C_1 value reliable? Explain your answer. Why does this situation happen? What parameter must be adjusted in the measurement to avoid this situation?

(10 pts)

- (b) Figure Q3.1 is the capacitance-voltage characteristic of a MOS structure on p-type substrate, measured at 1MHz. In this measurement, a MOS transistor is used and the source, the drain, and the substrate terminals are grounded. Is this result reasonable? Explain. (15 pts)

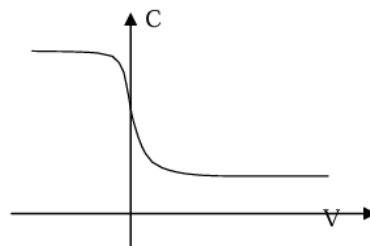
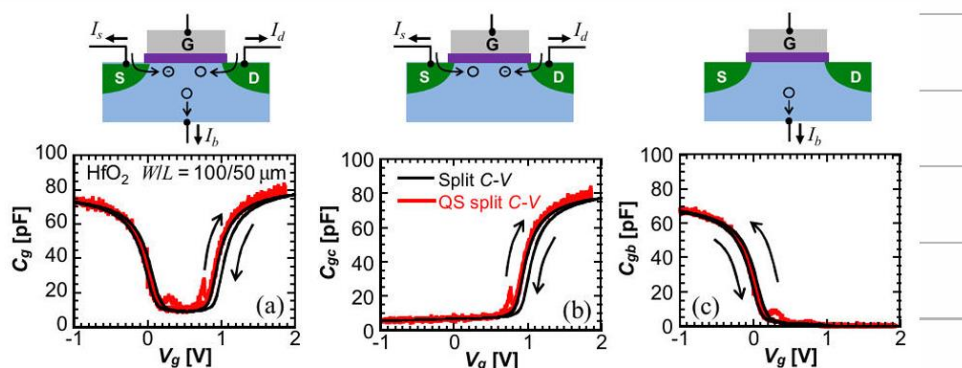


Fig. Q3.1

(a) Ans: His evaluation is not reliable. In this device structure, when $V_G = 4$ V, it is surely inversion mode. But the fact that the capacitance C_1 was increasing with time at a given V_G means that it was not in an equilibrium state. Thus **deep depletion** happened. The depletion region was extended beyond the maximum width at the equilibrium. As time passing, the equilibrium is restored inside the device through the generation of minority carrier in the near surface region, and the depletion region also is restored to the maximum width at the equilibrium. Thus total capacitance is increasing.

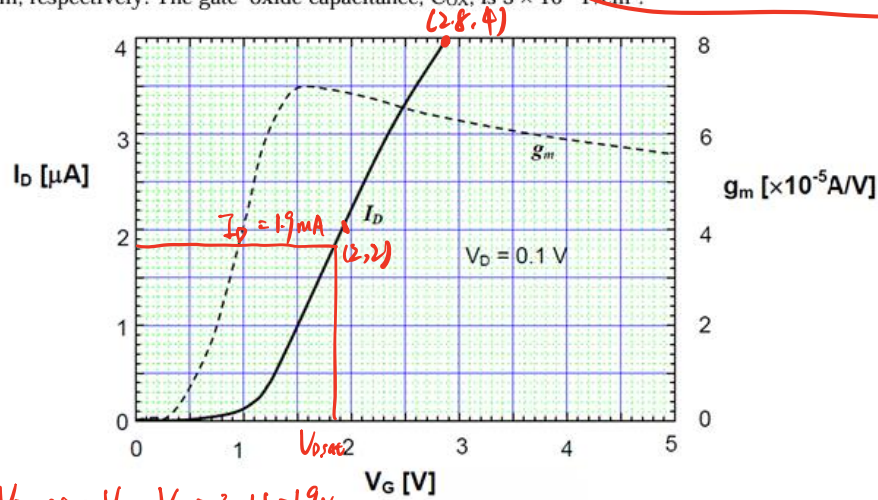
This phenomenon happens when DC bias sweep speed is too fast compared to the minority carrier generation rate. To avoid this deep depletion, the DC bias sweep speed must be lowered.

(b) The C-V curve is a typical high frequency C-V curve of MOS capacitor structure. The inversion capacitance is much lower than accumulation capacitance because the thermal generation of inversion charge cannot follow the high frequency signal of 1 MHz. But, in this case, the capacitance was measured using MOS transistor with source/drain. Since the source/drain is grounded, the inversion charges are supplied from source/drain. Therefore, there is no problem for inversion charges to follow the high frequency signal of 1 MHz. Then inversion capacitance will be similar to accumulation capacitance. Therefore, the C-V curve must look like a low frequency C-V curve when a MOS capacitor structure is used. The result is not reasonable.



Question 4

Linear I_D (and transconductance g_m) vs. linear V_G curves of an n-channel MOSFET were measured as shown in Fig. Q2.1. The channel width (W) and the channel length (L) of the device are $10\text{ }\mu\text{m}$ and $2\text{ }\mu\text{m}$, respectively. The gate oxide capacitance, C_{ox} , is $3 \times 10^{-7}\text{ F/cm}^2$.



$$V_{D,sat} = V_G - V_T = 3 - 1.1 = 1.9\text{ V}$$

Fig. Q4.1

Linear I_D vs. linear V_D curves of the same MOSFET for several different V_G were measured as shown in Fig. Q2.2.

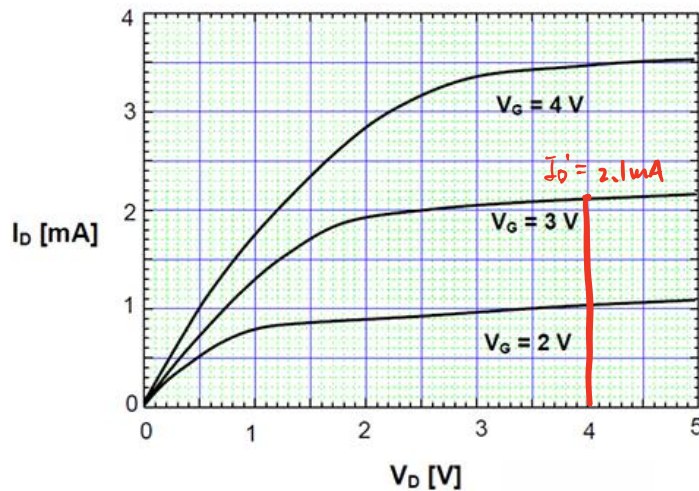


Fig. Q4.2

- Evaluate the value of maximum electron mobility. (10 pts)
- Please estimate the threshold voltage from Fig. Q2.1 by maximum transconductance method? (5 pts)
- The $I_D - V_D$ curves do not show the full saturation of I_D at high V_D due to channel length modulation. What is the distance between the drain edge and the channel pinch-off point under the bias condition of $V_D = 4.0\text{ V}$ and $V_G = 3\text{ V}$? (10 pts)

$$g_{m(max)} = \frac{4\mu A - 2\mu A}{2.8V - 2V} = 2.5\mu A/V$$

Q2. (a). 线性区: $I_D \approx \mu_n \frac{W}{L} C_{ox} (V_G - V_T) V_D$. V_D 很小时,

$$\Rightarrow g_m = \frac{\partial I_D}{\partial V_G} = \mu_n \frac{W}{L} C_{ox} \cdot V_D.$$

$$\Rightarrow \mu_{nmax} = \frac{g_{mmax}}{\frac{W}{L} C_{ox} \cdot V_D} = \frac{0.25 \times 10^{-5}}{(5 \times 3 \times 10^{-7}) \times 0.1} = \frac{50}{3} \text{ cm}^2/V \cdot s$$

实际 mosfet 的 μ_{eff} 基本在 $0 \sim 100 \text{ cm}^2/V \cdot s$

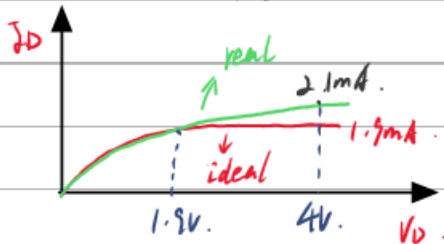
↳ 有效迁移率.

$$(b). V_T = V_{GS}(g_{mmax}) - \frac{I_D(g_{mmax})}{g_{mmax}} = 1.5 - \frac{1 \times 10^{-6}}{0.25 \times 10^{-5}} = 1.1V.$$

(c). from (b). $V_T = 1.1V$.

饱和条件: $V_D \geq V_G - V_T$

$\Rightarrow V_G = 3V, V_{D,sat} = 1.9V$. 此时 $I_D = 1.9mA$ (理论饱和 I)



由 $V_G = 3V, V_D = 4V$ 的图知.

此时 $I_D = 2.1mA$.

$$\text{Lecture 14. P40: } I_D' = \left(\frac{L}{L - \Delta L} \right) I_D \Rightarrow \frac{L}{L - \Delta L} = \frac{2.1}{1.9}.$$

$$\Rightarrow L - \Delta L = \frac{19}{21} \times 2\mu m = 1.8\mu m.$$

$$\therefore \Delta L = 0.19\mu m.$$

$$(b) \quad I_D(V_G) - I_D(V_{Gmax}) = g_{mmax} (V_G - V_{Gmax})$$

外推 $I_D = 0$ 时 $V_G = V_T$

$$\Rightarrow V_T = V_{Gmax} - \frac{I_D(V_{Gmax})}{g_{mmax}}$$

Question 5

Assume there is a MOS capacitor. The metal used in the capacitor is TaN, the oxide material is SiO₂, and the substrate is p-type Si. At 300 K, the doping concentration of the p-type substrate $N_A = 1.5 \times 10^{17} \text{ cm}^{-3}$. The fixed charge in the gate oxide Q_s is 10^{11} cm^{-2} , the work function of TaN is 4.02 eV, and the electron affinity of Si and SiO₂ are 4.01 eV and 9 eV, respectively. The MOS capacitor was measured by a semiconductor analyser. It was found the maximum capacitance is $5 \times 10^{-8} \text{ F/cm}^2$. Provided that, at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, $\epsilon_0/e = 5.52 \times 10^5 \text{ F cm}^{-1} \text{ C}^{-1}$, please calculate:

i. The gate oxide thickness t_{ox} and capacitance C_{ox} .

(10 pts)

ii. The flatband voltage V_{FB} .

(5 pts)

iii. The maximum depletion width X_d and the depletion charge density Q_d .

(20 pts)

iv. The threshold voltage V_T .

(5 pts)

v. If the gate metal TaN is replaced by a heavily n-type doped (n++) Si, what is the impact on the capacitance of the MOS device? Please discuss it qualitatively.

(10 pts)

(1) 栅氧化层厚度和栅氧化层单位面积电容 C_{ox} 。

(5 分)

C_{ox} 就是 C-V 测试中的最大电容值 $5 \times 10^{-8} \text{ F/cm}^2$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 5 \times 10^{-8} \text{ F/cm} \quad (2 \text{ 分})$$

$$t_{ox} = \frac{\epsilon_{ox}}{C_{ox}} = \frac{3.9 \times 8.854 \times 10^{-14}}{5 \times 10^{-8}} = 6.9 \times 10^{-6} \text{ cm} = 69 \text{ nm}$$

(3 分, 公式 1 分, 列出所有参数 1 分, 计算对了 1 分)

(2) 平带电压 V_{FB} 。

(5 分)

$$p_0 = n_i \exp\left[\frac{E_{Fi} - E_F}{kT}\right] = n_i \exp\left[\frac{(e\phi_b)}{kT}\right]$$

$$\Rightarrow \phi_b = E_{Fi} - E_F = \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right)$$

$$\phi_b = \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right) = 0.0259 \ln\left(\frac{1.5 \times 10^{17}}{1.5 \times 10^{10}}\right) = 0.418 \text{ V}$$

$$V_G = \phi_m - \phi_s = \phi_m - [\chi_s + E_G/2 + \phi_b]$$

$$= 4.02 - [4.01 + 0.56 + 0.418] = -0.97 \text{ V}$$

$$V_{FB} = \phi_{ms} - \frac{Q_s}{C_{ox}} = -0.97 - \frac{1.602 \times 10^{-19} \times 10^{11}}{5 \times 10^{-8}} = -1.29 \text{ V}$$

(3) 最大耗尽层宽度 x_d 和单位面积耗尽层电荷量 Q_d 。

(10 分)

$$x_d = \sqrt{\frac{4\epsilon_s \phi_b}{eN_a}} \quad (3 \text{ 分}) = \sqrt{\frac{4 \times 11.7 \times 8.854 \times 10^{-14} \times 0.418}{1.602 \times 10^{-19} \times 1.5 \times 10^{17}}} \quad (1 \text{ 分}) = 8.49 \times 10^{-6} \text{ cm} = 84.9 \text{ nm} \quad (1 \text{ 分})$$

$$Q_d = -eN_a x_d \quad (3 \text{ 分}) = -(1.602 \times 10^{-19}) \times 1.5 \times 10^{17} \times 8.49 \times 10^{-6} \quad (1 \text{ 分}) = -2.04 \times 10^{-7} \text{ C cm}^{-2} \quad (1 \text{ 分})$$

(4) 阈值电压 V_T 。

(5 分)

$$V_T = V_{FB} + \frac{2\sqrt{e\epsilon_s N_a \phi_b}}{C_{ox}} + 2\phi_b = V_{FB} + \frac{-Q_d}{C_{ox}} + 2\phi_b \quad (3 \text{ 分}) = -1.29 + \frac{2.04 \times 10^{-7}}{5 \times 10^{-8}} + 2 \times 0.418 = 3.63 \text{ V}$$

(2 分)

(5) 若将 TaN 换成功函数相同的 n 型多晶半导体材料, 会对 MOS 电容值产生怎样的影响? 请定性分析。

由于金属内存在大量自由电子, 所以当对金属加压时, 在金属和栅介质界面上, 不存在耗尽区 (2 分)。而当半导体材料作为栅极时, 尽管掺杂浓度很高, 但是仍然能够存在耗尽区在界面上 (2 分), 这将导致多晶硅耗尽电容和 C_{ox} 的串联 (1 分)。